

#This code will be used to demonstrate lemma 7.4. we demonstrate the existence of the curve $\gamma\beta$ in region R2.

#Construct the polynomial R from Theorem 7.1.2.

$$R := (2 \cdot w^3 \cdot z^3)^2 - (w^3 + z^3)^2 \cdot \left(z^2 + \frac{y \cdot x}{2} - \frac{y^2}{4} \right)^3$$

$$R := 4 w^6 z^6 - (w^3 + z^3)^2 \left(z^2 + \frac{1}{2} y x - \frac{1}{4} y^2 \right)^3 \quad (1)$$

#Apply the rational parametrization constructed in this chapter 4..

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify(numer(expand(subs(parametrization, R)))) ;

denom(expand(subs(parametrization, R))) ;

$$\begin{aligned} & - (a^{22} b^{10} + (-6 b^{11} + 8 b^9) a^{21} + (15 b^{12} + 6 b^{10} + 30 b^8) a^{20} + (-18 b^{13} - 102 b^{11} + 162 b^9 \\ & + 74 b^7) a^{19} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{18} + (24 b^{15} - 96 b^{13} \\ & - 762 b^{11} + 1884 b^9 + 1380 b^7 + 252 b^5) a^{17} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} \\ & + 1908 b^8 + 1092 b^6 + 145 b^4) a^{16} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 \\ & + 4692 b^7 + 1380 b^5 + 74 b^3) a^{15} + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 \\ & - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^{14} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\ & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{13} + (15 b^{20} + 198 b^{18} \\ & - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) \\ & a^{12} - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 \\ & + 127 b^4 + 17 b^2 + 1) a^{11} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} \\ & - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{10} + (8 b^{21} + 162 b^{19} + 1884 b^{17} \\ & + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^9 + (30 b^{20} \\ & + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^8 \\ & + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^7 \\ & + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^6 + (252 b^{17} \\ & + 1380 b^{15} + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^5 + (145 b^{16} + 600 b^{14} + 93 b^{12} \end{aligned}$$

$$\begin{aligned}
& + 198 b^{10} + 3 b^8) a^4 + (74 b^{15} + 162 b^{13} - 102 b^{11} - 18 b^9) a^3 + (30 b^{14} + 6 b^{12} \\
& + 15 b^{10}) a^2 + (8 b^{13} - 6 b^{11}) a + b^{12}) (a b - 1)^2 \\
& 64 a^{12} b^{12}
\end{aligned} \tag{2}$$

#Define r3

$$\begin{aligned}
r3 := & (a^{22} b^{10} + (-6 b^{11} + 8 b^9) a^{21} + (15 b^{12} + 6 b^{10} + 30 b^8) a^{20} + (-18 b^{13} - 102 b^{11} \\
& + 162 b^9 + 74 b^7) a^{19} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{18} + (24 b^{15} - 96 b^{13} \\
& - 762 b^{11} + 1884 b^9 + 1380 b^7 + 252 b^5) a^{17} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} \\
& + 1908 b^8 + 1092 b^6 + 145 b^4) a^{16} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 \\
& + 4692 b^7 + 1380 b^5 + 74 b^3) a^{15} + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 \\
& - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^{14} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\
& - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{13} + (15 b^{20} + 198 b^{18} - 3162 b^{16} \\
& - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{12} - 6 b (b^{20} \\
& + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 \\
& + 1) a^{11} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 \\
& - 3162 b^4 + 198 b^2 + 15) a^{10} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} \\
& - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^9 + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} \\
& - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^8 + (74 b^{19} + 1380 b^{17} + 4692 b^{15} \\
& + 3340 b^{13} - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^7 + (145 b^{18} + 1092 b^{16} + 1908 b^{14} \\
& - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^6 + (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} \\
& - 96 b^9 + 24 b^7) a^5 + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^4 + (74 b^{15} + 162 b^{13} \\
& - 102 b^{11} - 18 b^9) a^3 + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^2 + (8 b^{13} - 6 b^{11}) a + b^{12}) :
\end{aligned}$$

#Plot the curve r3=0 in region R2.

$$r := a - b :$$

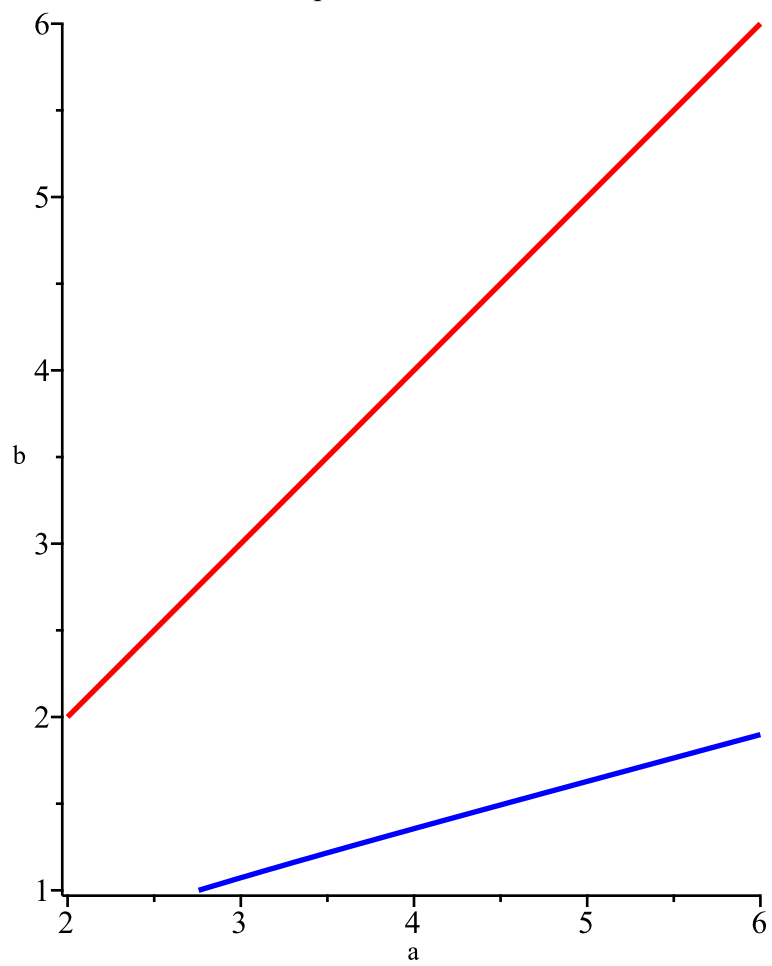
with(plots) :

```

implicitplot([r3=0, r=0], a=2 .. 6, b=1 .. (6),
  scaling=constrained,
  gridrefine=2,
  thickness=2,
  color=[blue, red],
  labels=["a", "b"],
  title="Graphic for r3=0 in A1");

```

Graphic for $r_3=0$ in A1



#Apply the diffeomorphism below between the region of the R2 parameters and the first quadrant constructing r3

transformation := {a = c + d + 1, b = d + 1} :

r3 := collect(expand(subs(transformation, r3)), c) :

#Compute the degree of r3 in c

degree(r3, c);

22

(3)

#Define the coefficients ci = coeff(r3, c, i)

seq(evalf(assign(cat('c', i), coeff(r3, c, i))), i = 0 .. 22);

#Compute the all degree of ci in d

seq(degree(eval(cat('c', i)), d), i = 0 .. 22);

32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10

(4)

#From here we will analyze the signal variation of all coefficients ci

seq(sign(evalf(coeff(c22, d, i))), i = 10 .. 0, -1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(5)

seq(sign(evalf(coeff(c21, d, i))), i = 11 .. 0, -1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(6)

seq(sign(evalf(coeff(c20, d, i))), i = 12 .. 0, -1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(7)

seq(sign(evalf(coeff(c19, d, i))), i = 13 .. 0, -1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(8)

seq(sign(evalf(coeff(c18, d, i))), i = 14 .. 0, -1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(9)

seq(sign(evalf(coeff(c17, d, i))), i = 15 .. 0, -1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(10)

seq(sign(evalf(coeff(c16, d, i))), i = 16 .. 0, -1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(11)

seq(sign(evalf(coeff(c15, d, i))), i = 17 .. 0, -1);

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(12)

seq(sign(evalf(coeff(c14, d, i))), i = 18 .. 0, -1);

$$\begin{aligned} &fsolve(c14=0, d, 0 \dots \text{infinity}); \\ &\quad -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ &\quad \quad \quad 4.128609601 \end{aligned} \tag{13}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c13, d, i))), i = 19 \dots 0, -1); \\ & fsolve(c13 = 0, d, 0 \dots \text{infinity}); \\ & \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & \quad \quad \quad 1.367483800 \end{aligned} \tag{14}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c12, d, i)), i = 20 .. 0, -1); \\ & fsolve(c12 = 0, d, 0 .. infinity); \\ & \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1 \\ & \quad \quad \quad 0.6088076874 \end{aligned} \tag{15}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c11, d, i))), i = 21 .. 0, -1); \\ & fsolve(c11 = 0, d, 0 .. infinity); \\ & -1, 1, 1, \\ & \quad 1 \\ & \quad \quad \quad 0.2099991402 \end{aligned} \tag{16}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c10, d, i))), i = 22 .. 0, -1); \\ & -1, \\ & \quad -1, -1, -1 \end{aligned} \quad (17)$$

$$\begin{aligned} & seq(sign(evalf(coeff(c9, d, i))), i = 23 \dots 0, -1); \\ & -1, \\ & \quad -1, -1, -1, -1 \end{aligned} \quad (18)$$

$$\begin{aligned} & seq(sign(evalf(coeff(c8, d, i))), i = 24 .. 0, -1); \\ & -1, \\ & \quad -1, -1, -1, -1, -1 \end{aligned} \quad (19)$$

$$\begin{aligned} & seq(sign(evalf(coeff(c7, d, i))), i = 25 .. 0, -1); \\ & -1, \\ & \quad -1, -1, -1, -1, -1, -1 \end{aligned} \quad (20)$$

$$\begin{aligned} & seq(sign(evalf(coeff(c6, d, i))), i = 26 .. 0, -1); \\ & -1, \\ & \quad -1, -1, -1, -1, -1, -1, -1 \end{aligned} \quad (21)$$

$$\begin{aligned} & seq(sign(evalf(coeff(c5,d,i))), i=27..0,-1); \\ & -1, \quad (22) \\ & -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

$$\begin{aligned} & seq(sign(evalf(coeff(c4, d, i))), i = 28 .. 0, -1); \\ & -1, \\ & \quad -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned} \quad (23)$$

[illegible]

